IoT– Enabled Smart Greenhouse Monitoring and Control System

Post Graduate Diploma in Embedded Systems Design (PG-DESD)

in

Sunbeam Institute of Information Technology, Pune

A project Report

Submitted by

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ABSTRACT

The IoT Enabled Greenhouse Monitoring and Control System is designed to provide an efficient and automated solution for maintaining optimal environmental conditions required for plant growth. Traditional greenhouse management relies heavily on manual monitoring and control, which can lead to inefficient resource utilization and inconsistent crop yield. This project overcomes these limitations by integrating sensors, embedded systems, and Internet of Things (IoT) technology. The system continuously monitors key environmental parameters such as temperature, humidity, soil moisture, and light intensity using appropriate sensors. The collected data is processed by a microcontroller and transmitted to a cloud platform for real-time visualization and remote monitoring. Based on predefined threshold values, control mechanisms such as irrigation systems, ventilation fans, and lighting are automatically activated or deactivated to maintain optimal greenhouse conditions. The proposed system reduces human intervention, improves resource efficiency, and enhances crop productivity. It demonstrates the practical application of IoT and embedded system concepts in smart agriculture and provides a scalable, cost-effective solution for modern greenhouse monitoring and control.

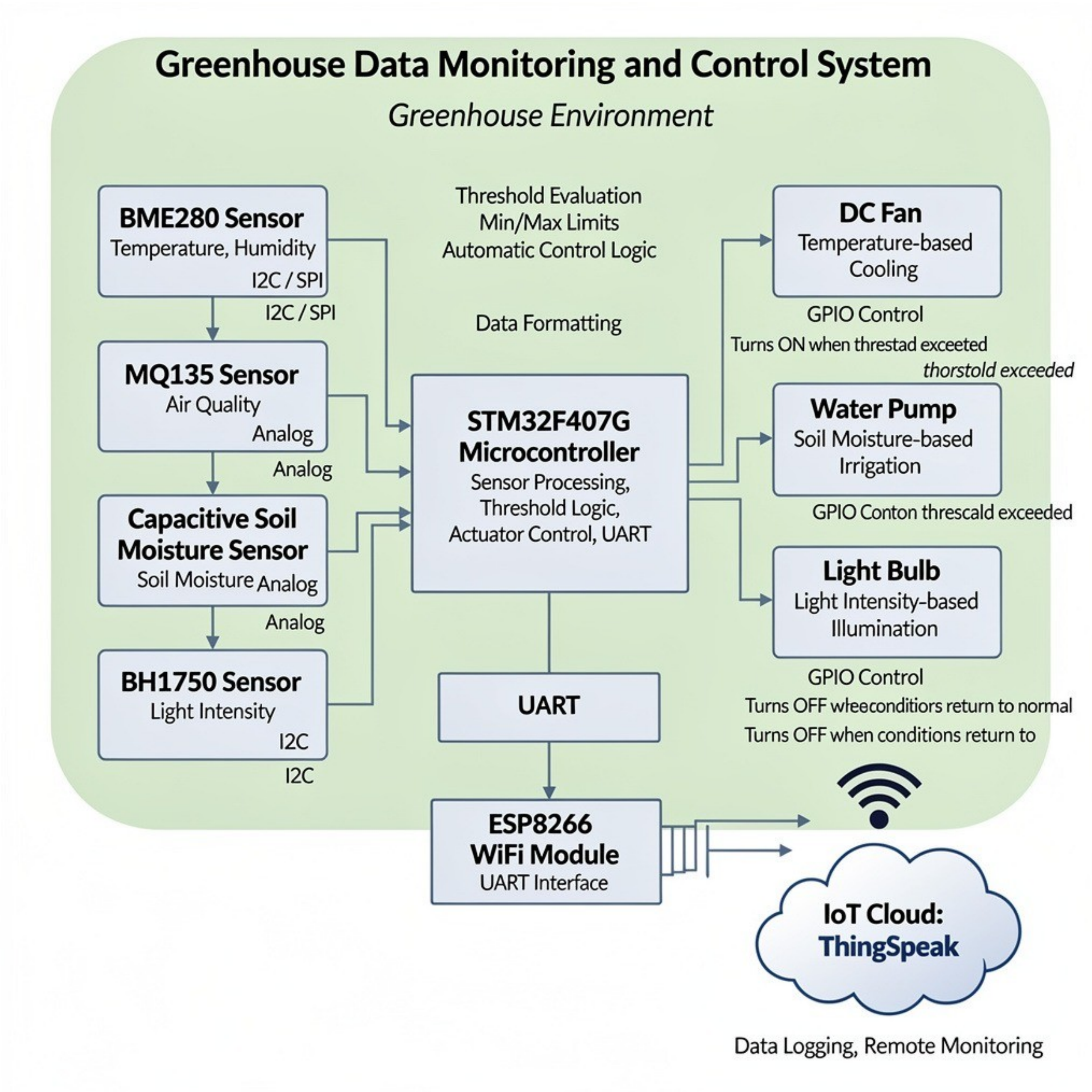
INTRODUCTION

Agriculture plays a vital role in ensuring food security, and greenhouse farming has emerged as an effective technique to improve crop yield by providing a controlled environment. However, traditional greenhouse management requires continuous manual monitoring of environmental parameters such as temperature, humidity, soil moisture, and light intensity, which is time-consuming and prone to human error.

The IoT Enabled Greenhouse Monitoring and Control System addresses these challenges by integrating sensors, microcontrollers, and cloud connectivity to enable real-time monitoring and automated control of greenhouse conditions. Using IoT technology, environmental data is continuously collected through sensors and transmitted to a cloud platform, allowing farmers or users to monitor conditions remotely. Based on predefined threshold values, control actions such as switching ON/OFF irrigation systems, ventilation, or lighting can be performed automatically or manually through a user interface.

This system improves resource utilization, reduces manual intervention, and enhances crop productivity by maintaining optimal growth conditions. The project demonstrates the practical application of embedded systems, sensors, and IoT platforms, making it suitable for real-world smart agriculture solutions.

BLOCK DIAGRAM



REQUIRED COMPONENTS

Sr.	COMPONENT NAME	QUANTITY
1.	MQ135 Gas Sensor	1
2.	BME280 temperature and Humidity Sensor	1
3.	BH1750 Light Intensity Sensor	1
4.	STM32F407G-DISC1	1
5.	Irrigation Pump	1
6.	DC Fan	1
7.	Voltage Regulator	1
8.	LED	1
9.	ESP8266	1
10.	5V single channel relay	2
11.	9V battery	2

MQ135 GAS SENSOR

The MQ135 is a semiconductor-based gas sensor used for detecting ozone (O_3) and other oxidizing gases in the air. It operates on the principle of chemiresistive sensing, where the resistance of the sensing material changes in the presence of target gases. The MQ132 sensor provides an analog voltage output proportional to the gas concentration, which is read using the ADC of the STM32 microcontroller. Due to its sensitivity and simple interfacing, the sensor is commonly used in environmental monitoring and air quality applications.

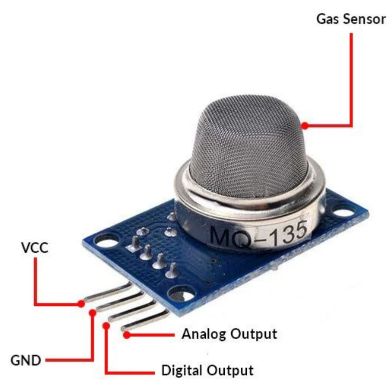


Fig : MQ135 Sensor

Commonly used in environmental monitoring and air quality applications.

- Sensor Type : Semiconductor gas sensor
- Target Gas : Ozone (O_3)
- Detection Principle : Change in resistance due to gas adsorption
- Operating Voltage : 5 V (heater and sensor circuit)
- Output Type : Analog voltage output
- Interface : ADC (Analog-to-Digital Converter)

- Detection Range : Approximately 10 ppb to 2 ppm (ozone)
- Sensitivity : High sensitivity to oxidizing gases
- Response Time : Fast response (few seconds)
- Preheat Time : Required (typically 24–48 hours for stable operation)
- Current Consumption : High due to internal heater (~150–200 mA)
- Operating Temperature : –10 °C to +50 °C (typical)
- Operating Humidity : 20 %RH to 90 %RH (non-condensing)

BH1750 LIGHT INTENSITY SENSOR

The BH1750 is a digital ambient light sensor that measures illuminance in lux (lx). It is widely used in embedded and IoT projects for applications such as automatic brightness control, street lighting, and display adjustment.

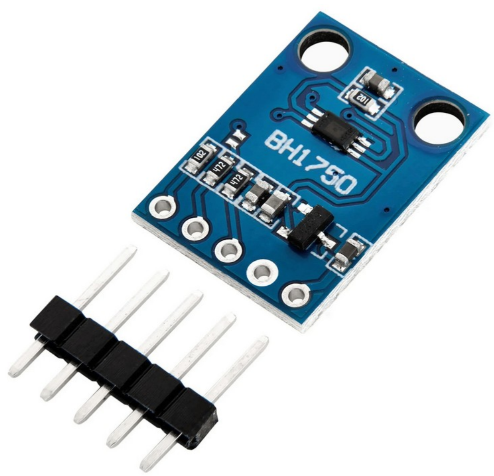


Fig: BH1750 sensor

Key Features

- Digital Output: Provides direct lux values via I2C interface; no ADC required.
- Wide Measurement Range: 1–65535 lux.

- High Resolution: Can measure light intensity up to 1 lx.
- Low Power Consumption: Suitable for battery-powered devices.
- Selectable Modes: Continuous or one-time measurements; high, low, and default resolution modes.
- Operating Voltage: 3.3–5V, compatible with microcontrollers like STM32, Arduino, and ESP32.

Technical Specifications

- Supply Voltage (VCC): 2.4V – 3.6V (typical 3.3V)
- Interface: I2C, 7-bit address: 0x23 or 0x5C (depending on ADDR pin)
- Measurement Accuracy: ±20% (typical)
- Response Time: ~120 ms in default mode
- Operating Temperature: -40°C to +80°C

BH1750 Operating Modes

The BH1750 has six primary measurement modes:

1. Continuous High-Resolution Mode (H-Resolution Mode, 1 lx)

- Measures 1 lx resolution continuously.
- Default mode after power-on.
- Typical measurement time: 120 ms.

2. Continuous High-Resolution Mode 2 (H-Resolution Mode2, 0.5 lx)

- Higher resolution than the default high-resolution mode.
- Continuous measurement at 0.5 lx resolution.
- Slower, typical measurement time: 120 ms.

3. Continuous Low-Resolution Mode (L-Resolution Mode, 4 lx)

- Fast measurement at 4 lx resolution.
- Continuous measurement mode.
- Typical measurement time: 16 ms.

4. One-Time High-Resolution Mode

- Measures once and then enters power-down mode.
- 1 lx resolution, typical measurement time: 120 ms.

5. One-Time High-Resolution Mode 2

- Measures once at 0.5 lx resolution, then enters power-down.

6. One-Time Low-Resolution Mode

- Measures once at 4 lx resolution, then enters power-down.
- Fastest mode: ~16 ms.

CAPACITIVE SOIL MOISTURE SENSOR

The capacitive soil moisture sensor is used to estimate soil water content by measuring the dielectric permittivity of the soil, which varies significantly with moisture level. Since water has a much higher dielectric constant than air or dry soil, changes in soil moisture result in measurable variations in capacitance. Unlike resistive soil moisture sensors, capacitive sensors do not use exposed metal electrodes, making them more stable, corrosion-resistant, and suitable for long-term deployment. The sensor provides an analog voltage output proportional to soil moisture level and is interfaced with the ADC of the STM32 microcontroller.

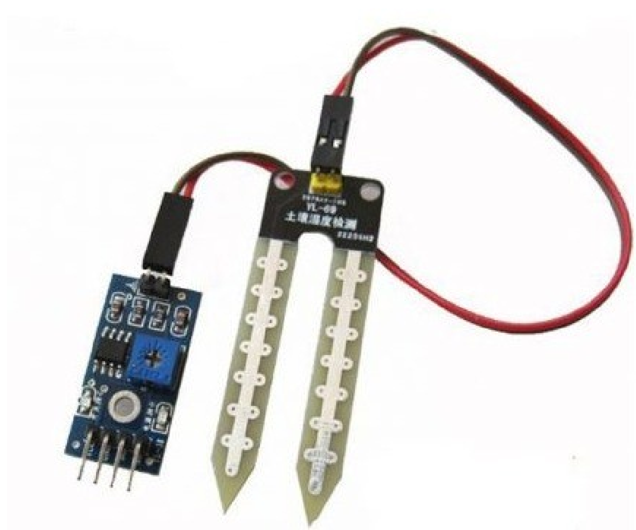


Fig : Soil Capacitive sensor

- Sensor Type : Capacitive soil moisture sensor
- Measurement Principle : Dielectric permittivity variation with soil moisture
- Measured Parameter : Soil moisture content (relative)
- Operating Voltage : 3.3 V to 5 V
- Output Type : Analog voltage output
- Interface : ADC (Analog-to-Digital Converter)
- Output Range : 0–3.0 V (typical, model dependent)
- Measurement Range : Dry to saturated soil (relative moisture level)
- Response Time : Fast (within a few seconds)
- Current Consumption : Approximately 5–20 mA
- Operating Temperature : 0 °C to 60 °C (typical)
- Electrode Design : No exposed metal electrodes (corrosion resistant)

BME280 SENSOR

The BME280 is a compact, low-power digital environmental sensor developed by Bosch Sensortec. It integrates three sensors in a single package to measure temperature, barometric pressure, and relative humidity. Due to its high accuracy, low power consumption, and digital interface support, the BME280 is widely used in embedded systems.

Fig : Bme280 SENSOR

- Sensor Type: Digital Temperature and Humidity sensor
- Temperature range: –40 °C to +85 °C
- Humidity range: 0 %RH to 100 %RH
- Temperature accuracy: ±1.0 °C
- Humidity accuracy: ±3 %RH
- Temperature resolution: 0.01 °C
- Humidity resolution: 0.008 %RH
- Sensor core voltage: 1.71 V to 3.6 V
- Interface voltage: 1.2 V to 3.6
- Communication Interface: Supports I^2C
- I^2C addresses: 0x76 or 0x77
- Temperature data: 20-bit
- Humidity data: 16-bit

Operating Modes:

The BME280 supports three operating modes:

Sleep Mode

- Default power-on mode
- No measurements performed
- Lowest power consumption
- Used to configure sensor registers

Forced Mode

- Sensor performs one measurement cycle and returns to sleep mode
- Suitable for low-power applications
- Measurement triggered by writing to control register

Normal Mode

- Continuous measurements at predefined intervals

- Uses internal standby time settings
- Suitable for real-time monitoring systems

STM32F407G-DISC1:

The STM32F407VG is a high-performance 32-bit ARM Cortex-M4 based microcontroller from STMicroelectronics, designed for real-time embedded and control applications. It operates at a maximum frequency of 168 MHz and includes a Floating Point Unit (FPU) and DSP instructions, making it suitable for signal processing, motor control, and automotive systems. The microcontroller provides rich peripheral support including CAN, SPI, I²C, UART, ADC, PWM timers, enabling easy interfacing with sensors, displays, and communication modules.

- Core : ARM Cortex-M4 with FPU and DSP
- Operating Frequency : Up to 168 MHz
- Operating Voltage : 1.8V to 3.6V (Typically 3.3V)
- Flash Memory : 1 MB
- SRAM : 192 KB
- CAN Interface : 2 × CAN controllers (CAN1 and CAN2)
- GPIO Pins : Up to 114 I/O pins
- Timers : Advanced, General-purpose, and Basic timers
- ADC : 3 × 12-bit ADC (up to 2.4 MSPS)
- Communication Interfaces : SPI, I²C, USART, USB OTG, Ethernet
- Package : LQFP-100 (used on development board)

ThingsBoard IoT Platform

ThingsBoard is an open-source Internet of Things (IoT) platform used for device management, data collection, processing, visualization, and monitoring. It enables users to collect data from IoT devices, store and analyze it, and visualize it through interactive dashboards. ThingsBoard supports multiple communication protocols and is widely used in smart agriculture, smart cities, industrial IoT, and environmental monitoring applications.

Key Features

- Real-time data collection and visualization
- Device and asset management
- Customizable dashboards and widgets
- Rule Engine for data processing and automation
- Alarm and notification support
- Scalability from small projects to enterprise-level systems
- Open-source with community and enterprise editions

Communication and Data Conversion Protocols

1. I²C (Inter-Integrated Circuit) Protocol

1.1 Introduction

I²C (Inter-Integrated Circuit) is a synchronous, serial communication protocol developed by Philips for short-distance communication between integrated circuits on the same PCB. It uses a **two-wire interface** and supports communication between one master and multiple slave devices.

1.2 I²C Bus Lines

- **SCL (Serial Clock Line):** Generates the clock signal, controlled by the master
- **SDA (Serial Data Line):** Used for bidirectional data transfer

Both lines are **open-drain**, requiring external pull-up resistors.

1.3 Addressing and Communication

Each slave device is identified by a **7-bit or 10-bit address**. Communication is initiated by the master using:

- **START condition**
- **Slave address + Read/Write bit**
- **ACK/NACK**
- **STOP condition**

1.4 Data Transfer

Data is transmitted in **8-bit frames**, followed by an acknowledgment bit. I²C supports:

- Standard mode (100 kHz)
- Fast mode (400 kHz)
- Fast mode plus (1 MHz)
- High-speed mode (3.4 MHz)

1.5 Advantages

- Requires only two wires

- Supports multiple devices on the same bus
- Simple hardware implementation

1.6 Applications

- Sensor interfacing (BME280, EEPROMs)
- RTC modules
- Embedded and IoT systems

2. ADC (Analog-to-Digital Converter)

2.1 Introduction

An ADC converts real-world **analog signals** such as voltage from sensors into **digital values** that can be processed by a microcontroller. ADCs are essential for interfacing analog sensors with digital embedded systems.

2.2 Working Principle

The ADC samples the analog input voltage and converts it into a digital number based on:

- **Reference voltage (Vref)**
- **Resolution**
- **Sampling rate**

2.3 Resolution

ADC resolution determines the smallest detectable voltage change:

- 8-bit ADC → 256 levels
- 10-bit ADC → 1024 levels
- 12-bit ADC → 4096 levels

2.4 ADC Conversion Formula

OutputDigital Output=(VrefVin)×(2n−1)

where *n* is the ADC resolution.

2.5 Types of ADC

- Successive Approximation ADC (SAR)
- Flash ADC

- Sigma-Delta ADC
- Dual-Slope ADC

2.6 Applications

- Reading temperature, soil moisture, and gas sensors
- Data acquisition systems
- Embedded control systems

3. UART (Universal Asynchronous Receiver Transmitter)

3.1 Introduction

UART is an asynchronous serial communication protocol used for full-duplex data transmission between devices. It does not require a clock signal and relies on predefined timing parameters.

3.2 UART Signals

- **TX (Transmit)**
- **RX (Receive)**
- **GND**

Optional signals:

- RTS, CTS (hardware flow control)

3.3 Data Frame Format

A typical UART data frame consists of:

- 1 Start bit
- 5–9 Data bits (usually 8)
- Optional Parity bit
- 1 or 2 Stop bits

3.4 Baud Rate

Baud rate defines the transmission speed (bits per second), e.g., 9600, 115200 bps. Both transmitter and receiver must use the same baud rate.

3.5 Advantages

- Simple and widely supported
- Full-duplex communication
- Low hardware complexity

3.6 Applications

- Debugging and logging
- Communication with GSM, GPS, Bluetooth modules

RESULT

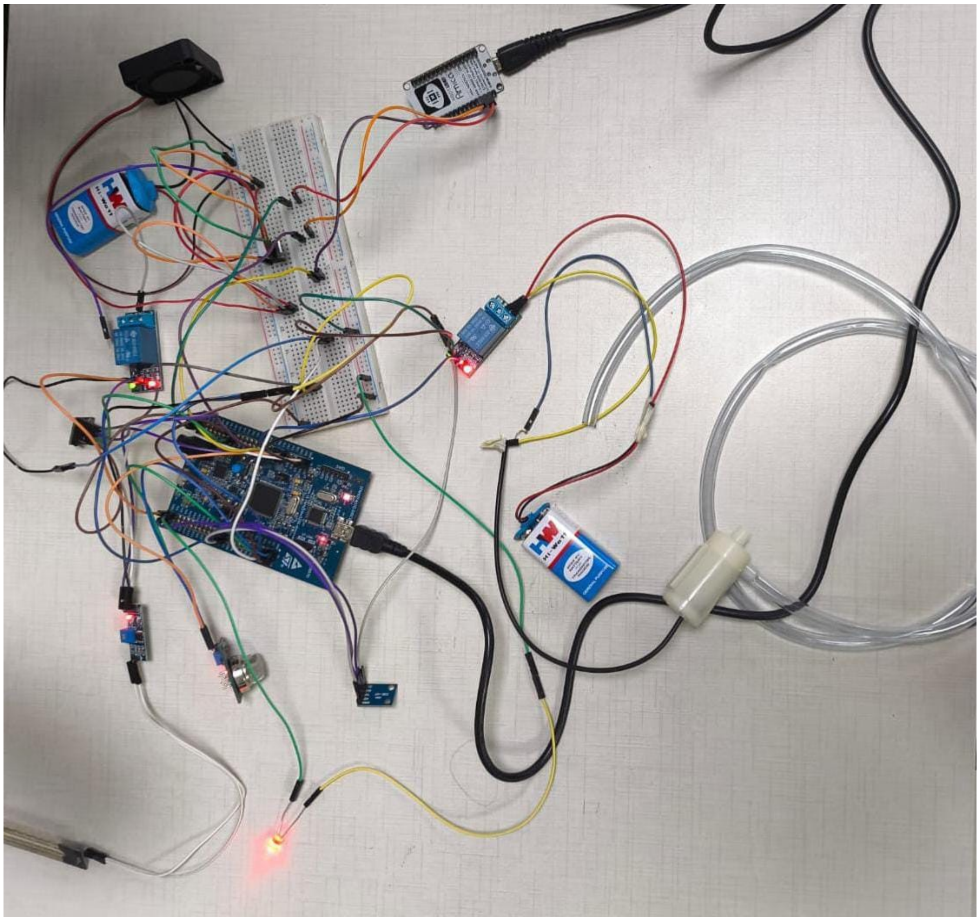


Fig1:Integration of all sensors to STM32F407G-DISC1

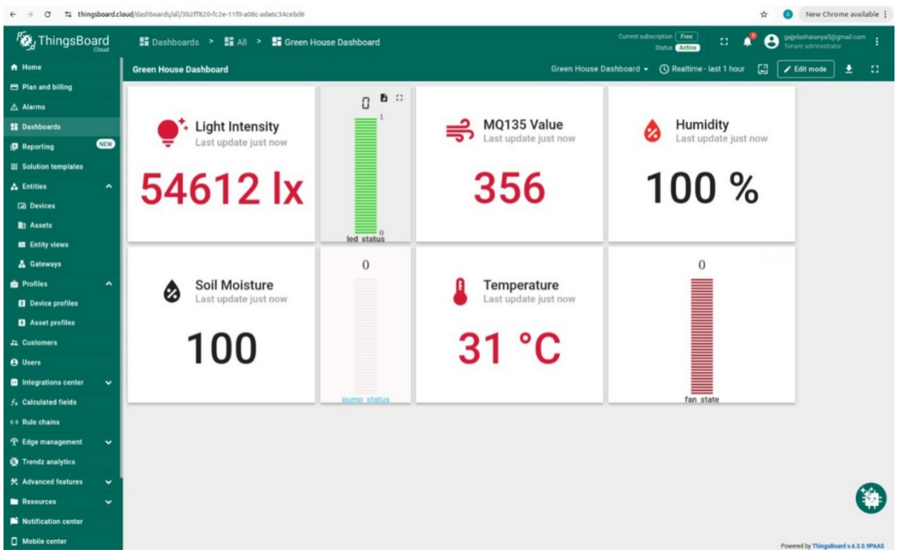


Fig2:Real-Time Monitoring Results from ThingsBoard

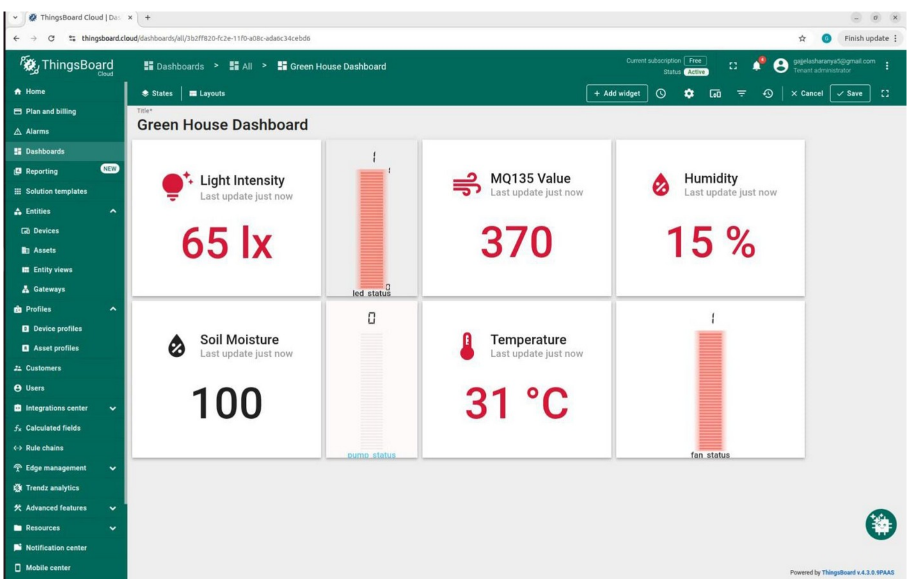


Fig3: Smart Greenhouse Monitoring Dashboard on IoT ThingsBoard

CONCLUSION

The IoT Enabled Greenhouse Monitoring and Control System successfully demonstrates an efficient and reliable method for monitoring environmental parameters and controlling greenhouse operations in real time. By integrating sensors with a microcontroller and cloud-based monitoring, the system provides accurate data visualization and automated control, leading to improved crop health and optimized resource usage. The implemented system reduces dependency on manual supervision, minimizes water and energy wastage, and enables remote access to greenhouse data. The project validates the effectiveness of IoT technology in modern agriculture and highlights how smart monitoring systems can significantly improve productivity and sustainability. Overall, the project meets its objectives and serves as a scalable solution for intelligent greenhouse management.

FUTURE SCOPE

The current system can be further enhanced and expanded in several ways:

- Integration of AI/ML algorithms for predictive analysis and intelligent decision-making based on historical data
- Addition of more sensors such as CO_2, pH, and nutrient sensors for advanced crop monitoring
- Implementation of a mobile application for real-time alerts and remote control
- Use of solar power to make the system energy efficient and suitable for remote areas
- Integration with automated fertigation and pesticide spraying systems
- Support for multiple greenhouses with centralized cloud monitoring
- Implementation of data analytics dashboards for long-term crop performance analysis

These enhancements can transform the system into a fully autonomous smart agriculture platform suitable for large-scale commercial greenhouse applications